Различные виды производящих ф-й

$$\tilde q = \tilde q(q,p,t)$$
$$\tilde p = \tilde p(q,p,t)$$

$2n + 2n$ перем.

1) q, p - независ.

$$\left(\sum_i \tilde p_i d\tilde q_i - \tilde H dt\right) - c\left(\sum_i p_i dq_i - H dt\right) = -dF$$

$$\sum_i \tilde p_i\left(\sum_j \frac{\partial \tilde q_i}{\partial q_j} dq_j + \sum_j \frac{\partial \tilde q_i}{\partial p_j} dp_j + \frac{\partial \tilde q_i}{\partial t} dt\right) - \tilde H dt = c\left(\sum_j p_j dq_j - H dt\right) - \sum_j \frac{\partial F}{\partial q_j} dq_j - \sum_j \frac{\partial F}{\partial p_j} dp_j - \frac{\partial F}{\partial t} dt$$

$dq_j: \quad \sum_i \tilde p_i \frac{\partial \tilde q_i}{\partial q_j} = c p_j - \frac{\partial F}{\partial q_j}$

$dp_j: \quad \sum_i \tilde p_i \frac{\partial \tilde q_i}{\partial p_j} = -\frac{\partial F}{\partial p_j}$

$dt: \quad \sum \tilde p_i \frac{\partial \tilde q_i}{\partial t} - \tilde H = -cH - \frac{\partial F}{\partial t}$

$$\frac{\partial F}{\partial q_j} = c p_j - \sum_i \tilde p_i \frac{\partial \tilde q_i}{\partial q_j}$$
$$\frac{\partial F}{\partial p_j} = -\sum_i \tilde p_i \frac{\partial \tilde q_i}{\partial p_j}$$
$$\tilde H = cH + \frac{\partial F}{\partial t} + \sum_i \tilde p_i \frac{\partial \tilde q_i}{\partial t}$$

1) $n=1,\ c=1,\ F=0$

$$\frac{\partial F}{\partial q} = p - \tilde p \frac{\partial \tilde q}{\partial q} = 0$$
$$\frac{\partial F}{\partial p} = -\tilde p \frac{\partial \tilde q}{\partial p} = 0$$
$$\frac{\partial \tilde q}{\partial p} = 0 \Rightarrow \tilde q = \tilde q(q,t)$$
$$\tilde p = p \Big/ \frac{\partial \tilde q}{\partial q}$$

2) $q, \tilde q$ - независ.

$$(*)\begin{cases} \tilde q = \tilde q(q,p,t) \\ \tilde p = \tilde p(q,p,t) \end{cases} \xRightarrow[\left|\frac{\partial \tilde q}{\partial p}\right| \neq 0]{} \begin{cases} p = p(q,\tilde q,t) \\ \tilde p = \tilde p(q,\tilde q,t) \end{cases}$$

Преобр. (*) - свобод. канонич., если

- канонич.
- $\left|\frac{\partial \tilde q}{\partial p}\right| \neq 0$

$\begin{cases} \tilde q = q \\ \tilde p = p \end{cases}$ не свобод. $\qquad \begin{cases} \tilde q = p \\ \tilde p = q \end{cases}$ - свобод.

$$\sum \tilde p_i d\tilde q_i - \tilde H dt = c\left(\sum p_i dq_i - H dt\right) - dS \overset{*}{=}$$

$$F(q,p,t) = F(q, p(q,\tilde q,t), t) = S(q,\tilde q,t)$$

$$\overset{*}{=} c\left(\sum p_i dq_i - H dt\right) - \sum \frac{\partial S}{\partial q_i} dq_i - \sum \frac{\partial S}{\partial \tilde q_i} d\tilde q_i - \frac{\partial S}{\partial t} dt$$

$d\tilde{q}_i: \quad \tilde{p}_i = -\frac{\partial S}{\partial \tilde{q}_i}$

$dq_i: \quad 0 = cp_i - \frac{\partial S}{\partial q_i}$

$dt: \quad -\tilde{H} = -cH - \frac{\partial S}{\partial t}$

Если $\left|\frac{\partial^2 S}{\partial q_i \partial \tilde{q}_i}\right| \neq 0$, то преобр. восстан.

$S = S(q, \tilde{q}, t)$ $\quad \left|\frac{\partial f}{\partial \tilde{q}}\right| \neq 0$

$\frac{\partial S}{\partial q_i} = f(q, \tilde{q}, t) = cp_i$

$\frac{\partial S}{\partial \tilde{q}_i} = g(q, \tilde{q}, t) = -\tilde{p}_i$

$$\left[\begin{array}{l} \begin{cases} p = p(\tilde{q}, \tilde{p}, t) \\ q = q(\tilde{q}, \tilde{p}, t) \end{cases} \\ \begin{cases} \tilde{p} = \tilde{p}(q, p, t) \\ \tilde{q} = \tilde{q}(q, p, t) \end{cases} \end{array}\right.$$

$\left|\frac{\partial^2 S}{\partial q \partial \tilde{q}}\right| = \left|\frac{\partial g}{\partial q}\right| \neq 0$

$\left|\frac{\partial^2 S}{\partial q \partial \tilde{q}}\right| = \left|\frac{\partial f}{\partial \tilde{q}}\right| \neq 0$

Другие типы критериев каноничности

1) Скобки Лагранжа

$\frac{\partial^2 F}{\partial q_k \partial q_j}: \quad -\sum_i \frac{\partial \tilde{p}_i}{\partial q_k}\frac{\partial \tilde{q}_i}{\partial q_j} - \cancel{\sum \tilde{p}_i \frac{\partial^2 \tilde{q}_i}{\partial q_k \partial q_j}} = -\sum_i \frac{\partial \tilde{p}_i}{\partial q_j}\frac{\partial \tilde{q}_i}{\partial q_k} - \cancel{\sum \tilde{p}_i \frac{\partial^2 \tilde{q}_i}{\partial q_j \partial q_k}}$ (продиф. обыч. крит.)

$A_{jk} = \sum_i \left(\frac{\partial \tilde{q}_i}{\partial q_j}\frac{\partial \tilde{p}_i}{\partial q_k} - \frac{\partial \tilde{q}_i}{\partial q_k}\frac{\partial \tilde{p}_i}{\partial q_j}\right) = 0$

$\frac{\partial^2 F}{\partial p_k \partial p_j}: \quad -\sum_i \frac{\partial \tilde{p}_i}{\partial p_k}\frac{\partial \tilde{q}_i}{\partial p_j} - \sum_i \tilde{p}_i \frac{\partial^2 \tilde{q}_i}{\partial p_k \partial p_j} = -\sum_i \frac{\partial \tilde{p}_i}{\partial p_j}\frac{\partial \tilde{q}_i}{\partial p_k} - \sum_i \tilde{p}_i \frac{\partial^2 \tilde{q}_i}{\partial p_j \partial p_k}$

$C_{jk} = \sum_i \left(\frac{\partial \tilde{q}_i}{\partial p_j}\frac{\partial \tilde{p}_i}{\partial p_k} - \frac{\partial \tilde{q}_i}{\partial p_k}\frac{\partial \tilde{p}_i}{\partial p_j}\right) = 0$

$\frac{\partial^2 F}{\partial p_k \partial q_j}: \quad c\delta_{jk} - \sum_i \frac{\partial \tilde{p}_i}{\partial p_k}\frac{\partial \tilde{q}_i}{\partial q_j} - \sum_i \tilde{p}_i \cancel{\frac{\partial^2 \tilde{q}_i}{\partial p_k \partial q_j}} = -\sum \frac{\partial \tilde{p}_i}{\partial q_j}\frac{\partial \tilde{q}_i}{\partial p_k} - \cancel{\sum \tilde{p}_i \frac{\partial^2 \tilde{q}}{\partial q_j \partial p_k}}$

$B_{jk} = \sum_i \frac{\partial \tilde{q}_i}{\partial q_j}\frac{\partial \tilde{p}_i}{\partial p_k} - \frac{\partial \tilde{q}_i}{\partial p_k}\frac{\partial \tilde{p}_i}{\partial q_j} = c\delta_{jk}$

Опр.

Скобки Лагранжа

$\varphi_i(x, y)$ – n штук $\qquad \psi_i(x, y)$ – n штук

$[x, y] = \sum_{i=1}^{n} \frac{\partial \varphi_i}{\partial x}\frac{\partial \psi_i}{\partial y} - \frac{\partial \varphi_i}{\partial y}\frac{\partial \psi_i}{\partial x}$

преобраз. канон. IFF $\exists c$: $\quad [q_j, q_k] = 0$

$[p_j, p_k] = 0$

$[q_j, p_k] = c\delta_{jk}$

2) $J = \begin{bmatrix} 0 & E_n \\ -E_n & 0 \end{bmatrix}$

симплектическая единица

$J^T = J^{-1} = -J$

$J^2 = -E_{2n}$

$$\left\|\begin{matrix} \|A_{jk}\| & \|B_{jk}\| \\ -\|B_{jk}\| & \|C_{jk}\| \end{matrix}\right\| = c\mathcal{J}$$

$$\mathcal{M}^T \mathcal{J} \mathcal{M} = \left\|\begin{matrix} \|A_{jk}\| & \|B_{jk}\| \\ -\|B_{jk}\| & \|C_{jk}\| \end{matrix}\right\|$$

$$\mathcal{M} = \begin{pmatrix} \frac{\partial \tilde{q}}{\partial q} & \frac{\partial \tilde{q}}{\partial p} \\ \frac{\partial \tilde{p}}{\partial q} & \frac{\partial \tilde{p}}{\partial p} \end{pmatrix}$$

преобр. канон. $\Leftrightarrow$ $\mathcal{M}^T \mathcal{J} \mathcal{M} = c\mathcal{J}$

3) Скобки Пуассона

$$\mathcal{M} \mathcal{J} \mathcal{M}^T = c\mathcal{J}$$

Δ. $\mathcal{M}^T \mathcal{J} \mathcal{M} = c\mathcal{J}$

$$\mathcal{M}^T \mathcal{J} \mathcal{M} \cdot \mathcal{J} \mathcal{M}^T = c\mathcal{J}\mathcal{J}\mathcal{M}^T = -c\mathcal{M}^T$$

$$\underbrace{(\mathcal{M}^T)^{-1} \mathcal{M}^T}_{E} \mathcal{J} \mathcal{M} \mathcal{J} \mathcal{M}^T = -c\,(\mathcal{M}^T)^{-1} \mathcal{M}^T = -cE$$

$$\underbrace{\mathcal{J}^{-1} \mathcal{J}}_{E} \mathcal{M} \mathcal{J} \mathcal{M}^T = -c\mathcal{J}^{-1} = c\mathcal{J}$$

$$\mathcal{M} \mathcal{J} \mathcal{M}^T = c\mathcal{J} \quad \blacktriangle.$$

$$(\tilde{q}_j, \tilde{q}_k) = 0 \qquad (\tilde{p}_j, \tilde{p}_k) = 0 \qquad (\tilde{q}_j, \tilde{p}_k) = c\,\delta_{jk}$$

Фазовый поток Гам. сист. как унивалентное канон. преобр.

H, q_0, p_0

$$\begin{cases} q = q(q_0, p_0, t) \\ p = p(q_0, p_0, t) \end{cases} \quad (*)$$

$$(q_0, p_0) \xrightarrow[\text{фазовый поток}]{} (q, p) \quad \text{при fix } t$$

th

Решения Гам. сист. (*) явл. унивалент. кан. преобр.

Δ. $\bar{z} = \begin{pmatrix} \bar{q} \\ \bar{p} \end{pmatrix}$ $\qquad H_z = \begin{pmatrix} \partial H/\partial q \\ \partial H/\partial p \end{pmatrix}$

$$\dot{z} = \mathcal{J} H_z \qquad \bar{z} = \bar{z}(\bar{z}_0, t)$$

$$\frac{d\mathcal{M}}{dt} = \mathcal{J} \frac{\partial H_z}{\partial \bar{z}_0^T}$$

$$\mathcal{M}^T \mathcal{J} \mathcal{M} = c\mathcal{J} = \mathcal{J} \quad \text{(нужно док-ть)}$$

при $t=0$: $\mathcal{M} = \left.\frac{\partial \bar{z}}{\partial \bar{z}_0^T}\right|_{t=0} = E$

выполнено

$$\frac{d}{dt}(\mathcal{M}^T \mathcal{J} \mathcal{M}) = \frac{d\mathcal{M}^T}{dt} \mathcal{J} \mathcal{M} + \mathcal{M}^T \mathcal{J} \frac{d\mathcal{M}}{dt} =$$

$$\frac{d\mathcal{M}}{dt} = \frac{d}{dt}\begin{Vmatrix} \frac{\partial z_1}{\partial z_1^0} & \cdots & \frac{\partial z_1}{\partial z_{2n}^0} \\ \vdots & & \vdots \\ \frac{\partial z_{2n}}{\partial z_1^0} & \cdots & \frac{\partial z_{2n}}{\partial z_{2n}^0} \end{Vmatrix} = \begin{Vmatrix} \frac{\partial \dot z_1}{\partial z_1^0} & \cdots & \frac{\partial \dot z_1}{\partial z_{2n}^0} \\ \vdots & & \vdots \\ \frac{\partial \dot z_{2n}}{\partial z_1^0} & \cdots & \frac{\partial \dot z_{2n}}{\partial z_{2n}^0} \end{Vmatrix} = \frac{\partial \dot{\bar z}}{\partial \bar z_0^T}$$

$$\frac{\partial H_z}{\partial \bar z_0^T} = \begin{Vmatrix} \frac{\partial^2 H}{\partial z_1^0 \partial z_1} & \cdots & \frac{\partial^2 H}{\partial z_{2n}^0 \partial z_1} \\ \cdots & & \cdots \\ \frac{\partial^2 H}{\partial z_1^0 \partial z_{2n}} & \cdots & \frac{\partial^2 H}{\partial z_{2n}^0 \partial z_{2n}} \end{Vmatrix} = \begin{Vmatrix} \sum_{k=1}^{2n} \frac{\partial^2 H}{\partial z_k \partial z_1}\frac{\partial z_k}{\partial z_1^0} & \cdots & \sum_{k=1}^{2n} \frac{\partial^2 H}{\partial z_k \partial z_1}\frac{\partial z_k}{\partial z_{2n}^0} \\ \cdots & & \\ \sum_{k=1}^{2n} \frac{\partial^2 H}{\partial z_k \partial z_{2n}}\frac{\partial z_k}{\partial z_1^0} & \cdots & \cdots \end{Vmatrix} = \begin{Vmatrix} \frac{\partial^2 H}{\partial z_1^2} & \cdots & \frac{\partial^2 H}{\partial z_1 \partial z_{2n}} \\ \vdots & & \vdots \\ \frac{\partial^2 H}{\partial z_{2n} \partial z_1} & \cdots & \frac{\partial^2 H}{\partial z_{2n}^2} \end{Vmatrix} \cdot$$

$$\cdot \begin{Vmatrix} \frac{\partial z_1}{\partial z_1^0} & \cdots & \frac{\partial z_1}{\partial z_{2n}^0} \\ \vdots & & \vdots \\ \frac{\partial z_{2n}}{\partial z_1^0} & \cdots & \frac{\partial z_{2n}}{\partial z_{2n}^0} \end{Vmatrix} = H_{zz}\mathcal{M}$$

$$\frac{d\mathcal{M}}{dt} = J\frac{H_z}{\partial \bar z_0^T} = JH_{zz}\mathcal{M}$$

$$\frac{d\mathcal{M}^T}{dt} = \mathcal{M}^T H_{zz}^T J^T$$

$$= \mathcal{M}^T H_{zz}^T \underbrace{J^T J}_{E} \mathcal{M} + \mathcal{M}^T \underbrace{J J}_{-E} H_{zz}\mathcal{M} = \mathcal{M}^T \underbrace{H_{zz}^T}\mathcal{M} - \mathcal{M}^T \underbrace{H_{zz}}\mathcal{M} = 0$$

симм. матр.

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